Bihar Engineering University, Patna

B.Tech. 7th Semester Examination, 2023

Course: B. Tech. *Time: 03 Hours*

Code: 101703 *Subject: Pavement Designs* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) In cement warping and frictional stresses in a cement concrete slab are 210N/mm², 290N/mm² and 10N/mm² respectively, the critical combination of stresses during summer midday is

(i) 290 N/mm² (ii) 390 N/mm² (iii) 490 N/mm² (iv) 590 N/mm²

(b) Which one of the following methods is used in the design of Rigid pavement?

(i) CBR method (ii) Group index method

(iii) Westergaard's method (iv) McLeod's method

(c) Which one of the following is not a desirable property of the subgrade soil as a highway material?

(i) Stability (ii) Ease of compaction

(iii) Good drainage (iv) Bitumen adhesion

(d) The general requirement in constructing a reinforced concrete road is to place a single layer of reinforcement

(i) Near the bottom of the slab (ii) Near the top of the slab

(iii) At the middle (iv) Equally distributed at the top and bottom

(e) As per latest IRC guidelines for designing flexible pavement by CBR method, the load parameter required is

(i) Number of commercial vehicles per day

(ii) Cumulative standard axle in msa

(iii) Equivalent single axle load

(iv) Number of vehicles during design life

(f) The longitudinal joints in cement concrete pavements are constructed with suitable

(i) Tie bars (ii) Small spacing

(iii) Torsion rods (iv) Shear rods

(g) Which one of the following is not the correct type of critical load position in pavement slab design for the load on the pavement surface?

(i) Interior loading (ii) Edge loading

(iii) Eccentric loading (iv) Corner loading

(h) Which one of the following binders is recommended for a wet and cold climate?

(i) 80/100 penetration asphalt (ii) Tar

(iii) Cutback (iv) Emulsion

(i) An ideal horizontal transition curve is a

(i) Parabola (ii) Circle

(iii) Clothoid spiral (iv) Hyperbola

(j) Light reflecting devices used to guide the driver along the proper alignment are called

(i) Rumble strips (ii) Delineators

(iii) Attenuators (iv) Litter bin

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Q.2 Explain type of joints in cement concrete pavement and their functions. [14]

Q.3 (a) A plate load test is carried out on submerged soil using a 300mm radius rigid plate. A load of 5Tons resulted in a deflection of 1.20mm. What is the elastic modulus of the soil by considering the poission's ratio as 0.50? [7]

(b) Difference between Flexible and Rigid Pavement. [7]

Q.4 (a) Consider the following data with respect to the design of flexible pavement: [4]
Design wheel load = 4200kg
Tyre Pressure = 6kg/cm^2
Elastic modulus = 150kg/cm^2
Permissible deflection = 0.25cm

(b) A sub grade soil has following properties, Soil passing 0.075mm sieve = 65%, LL= 52%, PL= 43%. Design pavement section by Group Index method for heavy traffic with 450 commercial vehicle per day. [10]

Q.5 (a) What are the different layers of flexible pavement? Explain the function of each layer. [10]

(b) Explain the assumptions of Boussinesq's theory. [4]

Q.6 (a) What is the critical thickness of a prestressed concrete pavement using westergaard's Corner load formula to support a maximum wheel load of 4200Kg? Allow 10% for impact. Tyre pressure may be taken as 7kg/cm^2. Assume flexural strength of concrete as 50kg/cm^2, FOS as 2, subgrade reaction for plastic mix road as 6kg/cm^2, and modulus of elasticity as $3*10^5$kg/cm^2. [7]

(b) What is the deflection at the surface of a flexible pavement due to wheel load of 40kN and a tyre pressure of 0.5MPa? The value of E for pavement and subgrade is 20MPa. [7]

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Q.7 What are the various methods of flexible pavements design. Explain the details of each method. [14]

Q.8 Design the Reinforcement for a 15cm thick, simply supported RCC pavement. The contraction joints are spaced at 12m intervals. The pavement width is 7.5m, comprising of two lanes. Allowable stress for steel is 2800kg/cm^2. Take f =1.5. [14]

Q.9 Write short notes on ***any two*** of the following: [7x2=14]

(a) Warping Stress
(b) Design Wheel load and Contact Pressure
(c) Type of flexible Pavement Failure
(d) Radius of Relative Stiffness

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